DEPARTMENT OF MATHEMATICS, IIT MADRAS

MA5950 — Mathematical Finance: Theory & Implementation

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Lecture 1: Financial Markets Overview

Bonds · Annuities · Equities · Derivatives

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Today's Agenda

1. What *is* a financial market, really?
2. **Bonds** – the formal IOU
3. **Annuities** – a salary in reverse
4. **Equities** – ownership, not lending
5. **Derivatives** – contracts on contracts
6. Why this map matters for the rest of the course

Learning Objectives

By the end of this lecture, you will be able to:

- Explain what a financial market does, in one sentence.
- Read a bond's term sheet: face value, coupon, maturity, yield.
- State the defining risk difference between a fixed claim and a residual claim.
- Name the four basic derivative contract types and sketch a payoff diagram.

What Is a Financial Market?

The Marketplace for Time and Risk

A vegetable market matches a farmer's tomatoes with a household's demand for tomatoes *today*. A financial market matches something less tangible: people who want to move **money across time**, and people who want to move **risk** to whoever can bear it best. The "price" that clears this market is the interest rate.

- A young engineer has spare income *today* but needs a pension in 40 years $\rightarrow$ wants to move money *forward*.
- A company needs cash *today* to build a factory, repays from future profits $\rightarrow$ wants to move money *backward*.

Financial Market: Definition

Definition

A **financial market** is any organized venue or mechanism through which financial claims are issued and traded between savers, borrowers, and risk-transferring parties, at prices set by supply and demand.

Primary Market

A claim is created and sold *for the first time*.
e.g. IPO, a new G-Sec auction.

Secondary Market

Existing claims change hands between investors.
e.g. trading on NSE/BSE.

A Map of Asset Classes

Asset class	What you own	Nature of claim
Bonds	Creditor's claim on issuer	**Fixed**
Annuities	Stream of periodic payments	**Fixed / contingent**
Equities	Ownership stake	**Residual, variable**
Derivatives	Contract on another asset	Payoff *derived* from underlying

The most important line above: fixed vs. residual.

Bonds: A Formal IOU

A Bond Is a Formal IOU

Lend a friend Rs. 100; she promises Rs. 110 back in a year. That handshake deal *is*, economically, a bond. A real bond just formalizes it: standardized terms, made tradable, so thousands of "friends" can lend to one large borrower.

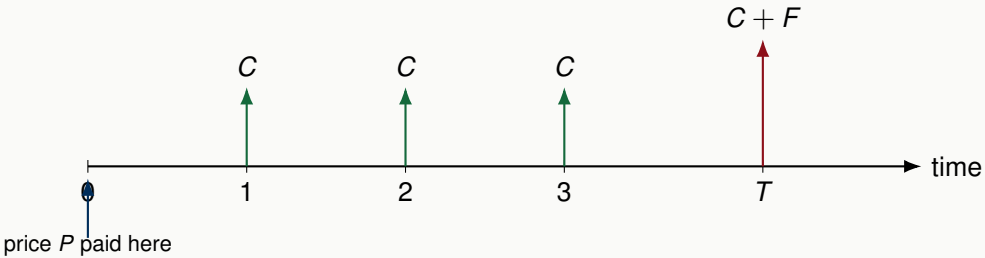
Bond: Definition

Definition

A **bond** is a contract in which the issuer promises the holder:

- a series of periodic **coupon** payments C, and
- a lump-sum **face value** F at **maturity** T.

Anatomy of a Bond – Cash Flow Timeline



- **Zero-coupon bond**: $C = 0$; bought at a discount, matures at F. The building block for the yield curve (Lecture 3 & the Lab).

Key Bond Terminology

- **Face value** F: repaid at maturity (typically Rs. 100 or Rs. 1000).
- **Coupon rate** c: contractual annual rate; $C = c \cdot F/m$ if paid m times a year.
- **Maturity** T: date the bond ceases to exist.
- **Yield to maturity (YTM)**: the single rate equating PV of all cash flows to price P. *(Full derivation next lecture.)*

Example: Reading a Bond Term Sheet

Government of India Bond

$F =$ Rs. 100, annual coupon 7%, matures in 3 years.

Cash flows:

- End of year 1: Rs. 7
- End of year 2: Rs. 7
- End of year 3: Rs. 107 (coupon + face value)

Next lecture: turn this list into a fair price P.

Why Bond Prices and Yields Move Oppositely

Remark

A bond's coupons are **fixed** once issued. If market rates rise, new bonds offer higher coupons $\rightarrow$ your old bond looks less attractive $\rightarrow$ its *price* must fall so its *yield* catches up.

$$\uparrow \text{rates} \iff \downarrow \text{bond prices}$$

Government vs. Corporate Bonds

- **Government bonds** (G-Secs, Treasuries): treated as (near) **default-free**. Only question: *how much to discount, purely for time*.
- **Corporate bonds**: add a second layer – **credit risk**, the chance of non-payment.

We build our machinery on government bonds first, precisely to isolate the "pure time value" problem.

Annuities: A Salary in Reverse

An Annuity Is a Salary in Reverse

At work: labour today $\rightarrow$ salary at intervals. An annuity flips it: hand over a lump sum *today* $\rightarrow$ receive a fixed "salary" at regular intervals later. It answers: *"I have savings now – how do I turn them into steady income later?"*

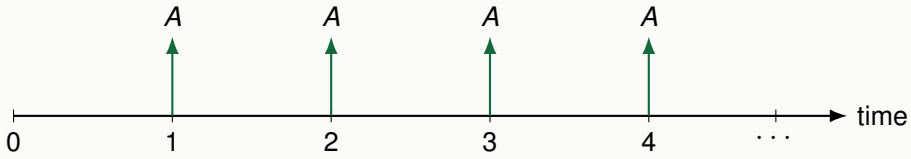
Annuity: Definition and Types

Definition

An **annuity** is a series of equal payments made at fixed intervals over a specified length of time.

- **Ordinary annuity**: payments at the *end* of each period.
- **Annuity due**: payments at the *start* of each period.
- **Perpetuity**: no end date – payments continue forever.

Ordinary Annuity – Timeline



$$PV = A \cdot \frac{1 - (1+r)^{-n}}{r} \quad \xrightarrow{n \to \infty} \quad PV = \frac{A}{r}$$

(We derive this from scratch next lecture – keep it in the back of your mind.)

Annuities in Everyday Life

Recognize the pattern

- A home loan EMI: a bond to the *bank*, an annuity you *owe*.
- A retirement pension plan.
- A scholarship disbursed monthly.
- EPFO pension payouts.

Equities: Landlord vs. Business Partner

Landlord vs. Business Partner

A bondholder is a **landlord**: fixed rent (coupon), regardless of the tenant's fortunes. An equity holder is a **business partner**: no fixed payment – you share the profit *or* the loss, in proportion to your stake. This is a **residual claim**: paid only after every fixed obligation is settled.

Equity: Definition

Definition

Equity represents fractional ownership of a company, entitling the holder to:

- a pro-rata claim on residual profits (dividends, retained earnings),
- a pro-rata claim on residual assets in liquidation,
- typically, voting rights on major decisions.

Bonds vs. Equities: The Core Trade-off

	Bond	**Equity**
Claim	Fixed, contractual	Residual, variable
Bankruptcy priority	Senior	Junior
Upside	Capped	Unlimited
Downside	Limited (until default)	Can go to zero
Volatility	Lower	Higher

This fixed-vs-residual spectrum recurs all term: utility theory, CAPM, portfolios.

Derivatives: An Insurance Policy on a Price

A Derivative Is an Insurance Policy on a Price

A wheat farmer fears prices *falling*; a flour mill fears prices *rising*. Today, they agree: "In 3 months, deliver 100 tonnes at Rs. 2000/tonne, whatever the market says then." Neither owns wheat right now under this contract – its value is entirely *derived* from the future price of wheat.

Derivative: Definition and the Four Families

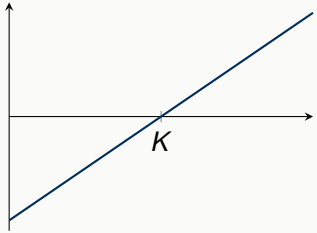
Definition

A **derivative** is a contract whose value is derived from an underlying asset, rate, or index.

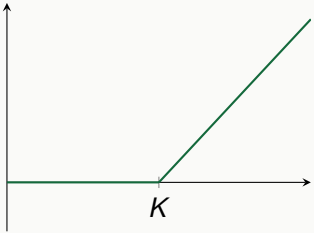
- **Forward**: custom OTC agreement, fixed price, future date.
- **Future**: standardized, exchange-traded forward; clearinghouse-guaranteed.
- **Option**: *right*, not obligation, to buy (call) / sell (put) at strike K – pay a premium, like insurance.
- **Swap**: exchange one cash-flow stream for another (e.g., fixed for floating rate).

Payoff Diagrams

Long Forward



Long Call



The call's payoff is *kinked* at K: flat below, rising above. That "floor at zero" is what the premium buys.

Why Hedge Instead of Just Waiting?

The mill's dilemma

The mill could wait 3 months and buy wheat at spot. But its *own* pricing to bakeries may depend on knowing wheat cost *in advance*.

A forward doesn't raise average profit – it trades away upside for a guaranteed floor on downside.

That is the essence of hedging.

Key Takeaways

Key Takeaway

- Financial markets match money across **time** and transfer **risk**.
- **Bond** = formal IOU: fixed coupons + face value. Creditor claim, senior, capped.
- **Annuity** = salary in reverse; priced with the discounting tools of Lecture 2.
- **Equity** = ownership; residual, uncapped claim – the polar opposite of a bond.
- **Derivative** = contract whose value is *derived* from an underlying; used to hedge or speculate.

Check Your Understanding

Discuss with your neighbor (2 minutes)

1. Name two risks a bond investor faces, even holding a government bond.
2. Classify: a fixed deposit; a partnership stake; a rainfall-indexed crop insurance payout.
3. Why must a bond's price fall when market rates rise? (No formulas – intuition only.)

Looking Ahead

Every formula previewed today needs one question answered first:

How much is Rs. 1 tomorrow worth today?

That is the **Time Value of Money** – Lecture 2.